This documentation and process conversion measures necessary to comply with this revision shall be completed by 22 Oct 94.

INCH-POUND

MIL-S-19500/420D
22 July 1994
SUPERSEDING
MIL-S-19500/420C
15 June 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER, RECTIFIER, GENERAL PURPOSE
TYPES 1N5550 THROUGH 1N5554, 1N5550US THROUGH 1N5554US
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, general purpose, semiconductor diodes. Four levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to DO-41), figure 2, and figures 3, 4, and 5 for JANHC AND JANKC.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C and ratings apply to all case outlines.

Type	$V_{(BR)min}$	V_{RWM}	I_{O1} T_L = +55°C; L = .375 inch 1/ 2/ 3/	I_{FSM} I_O = 2 A dc t_p = 1/120 s T_A = +55°C	T_{op}, T_{STG} (T_J)	I_{O2} T_A = +55°C 2/ 4/
	V dc	V dc	A dc	A(pk)	°C	A dc
1N5550, 1N5550US	240	200	5	100	-65 to +175	3
1N5551, 1N5551US	460	400	5	100	-65 to +175	3
1N5552, 1N5552US	660	600	5	100	-65 to +175	3
1N5553, 1N5553US	880	800	5	100	-65 to +175	3
1N5554, 1N5554US	1100	1,000	5	100	-65 to +175	3

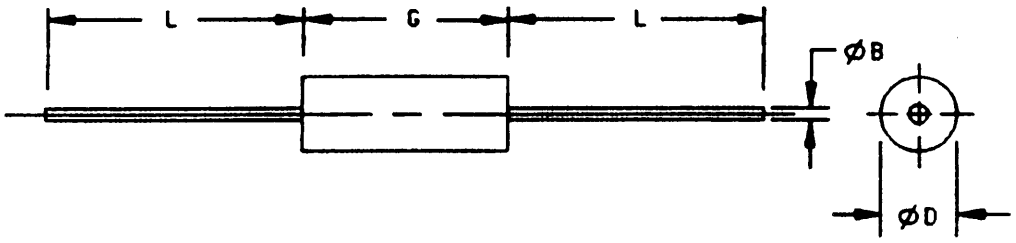
1/ Derate linearly at 41.6 mA/°C above T_L = +55°C at L = .375 inch (see 6.4).
2/ An I_O of up to 6 A dc is allowable provided that appropriate heat sinking or forced air cooling maintains the maximum junction temperature at or below +175°C as proven by the junction temperature rise test (see 6.4). Barometric pressure reduced:
1N5550, 1N5551, 1N5552 - 8 mmHg (100,000 feet).
1N5553, 1N5554 - 33 mmHg (70,000 feet).
3/ Does not apply to surface mount devices.
4/ Derate linearly at 25 mA/°C above T_A = +55°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 311.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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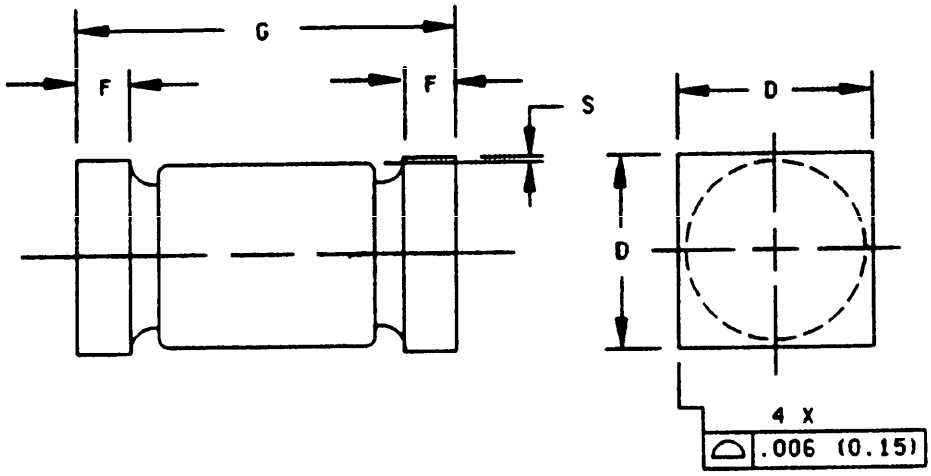


Dimensions					Notes
Ltr	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	.037	.042	0.94	1.07	
ØD	.115	.180	2.92	4.57	3,4
G	.130	.300	3.30	7.62	3
L	.900	1.300	22.86	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions G and ØD include all components of the diode periphery except the sections of leads over which the diameter is controlled.
4. Dimension ØD shall be measured at the largest diameter.

FIGURE 1. Semiconductor device, diode 1N5550 through 1N5554, JAN, JANTX, JANTXV, and JANS.

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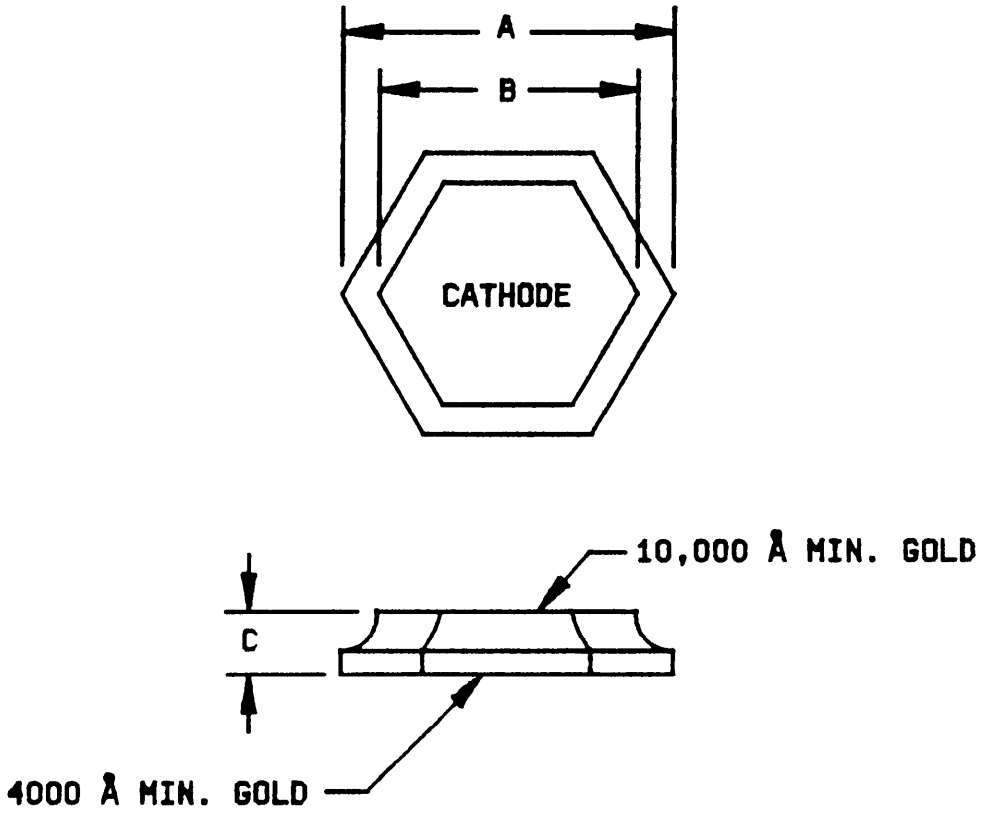


Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
G	.200	.275	5.08	6.99
F	.019	.034	0.48	0.86
S	.003		0.08	
D	.137	.180	3.48	4.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions of 1N5550US through 1N5554US.

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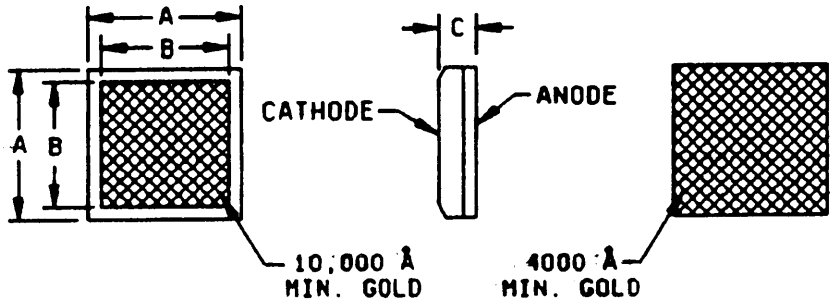
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.085	.091	2.16	2.31
B	.072	.078	1.83	1.98
C	.008	.014	0.20	0.36

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. JANHCA and JANKCA (A-version) die dimensions.

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Method	Conditions

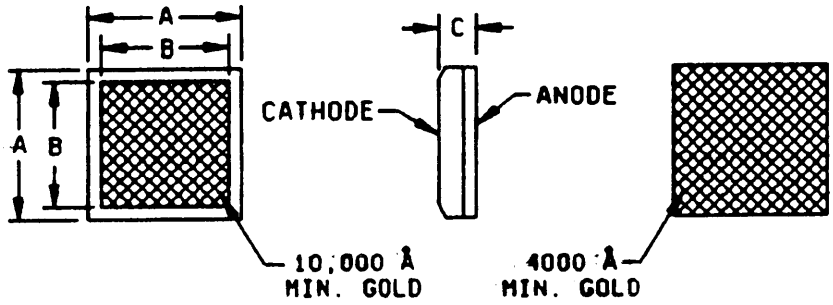


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.088	.092	2.24	2.34
B	.070	.077	1.78	1.96
C	.007	.035	0.18	0.89

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. JANHCB (B-version) die dimensions.

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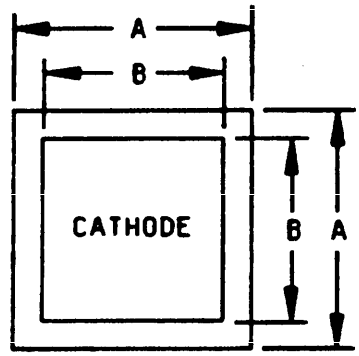


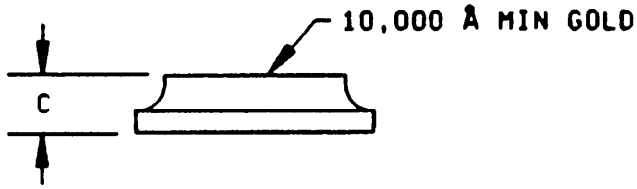
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.088	.092	2.24	2.34
B	.070	.077	1.78	1.96
C	.007	.035	0.18	0.89

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. JANHCB (B-version) die dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.060	.065	1.52	1.65
B	.052	.058	1.32	1.47
C	.008	.014	0.20	0.36

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. JANHCC (C-version) die dimensions.

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4.5.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.3 Scope display evaluation. Scope display evaluation shall be in accordance with test method 4023 of MIL-STD-750.

4.5.4 Junction temperature rise test. This test shall be performed in accordance with 4.5.4.1 and figures 6, 7, and 8.

4.5.4.1 Test procedure. Determine the temperature coefficient of forward voltage (TCV_F) using a lower reference temperature, (T_1) of 25°C to 30°C, an upper reference temperature (T_2) of 150°C to 200°C, and a reference current (I_{REF}) of 1 to 10 mA dc.

$$TCV_F = \frac{V_{F1} - V_{F2}}{T_2 - T_1}$$

Where $V_{F1} = V_F$ at I_{REF} at T_1 and

$V_{F2} = V_F$ at I_{REF} at T_2

At the option of the manufacturer, an average TCV_F based on a random sample of at least 25 production rectifiers may be used in the determination of ΔT_J. The diode shall be mounted at the specified lead length, and with S1 open ($I_O = 0$) read $V_{FO} = V_F$ at I_{REF}. Close S1 and adjust the power source and R_L for the specified I_O. After thermal equilibrium has been established at the specified lead temperature (T_L), read $V_{F3} = V_F$ at I_{REF} at 250 ± 150 µs after the power pulse. I_{REF} shall be within ±1 percent of the value used for calibration. Forced moving air or draft shall not be permitted across the devices during test.

$$\Delta T_J = \frac{V_{FO} - V_{F3}}{TCV_F}$$

4.5.5 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3101 or 4081 of MIL-STD-750. Read and record data in accordance with group E herein and shall be included in the qualification report. Forced moving air or draft shall not be permitted across the devices during test. The maximum limit for $R_{\theta JL}$ under these test condition shall be $R_{\theta JL} \leq 22°C/W$ for L = .375; $R_{\theta JL} \leq 11°C/W$ for L= 0 (US version). The following conditions shall apply:

a. I_H . . . . . 2 A minimum.

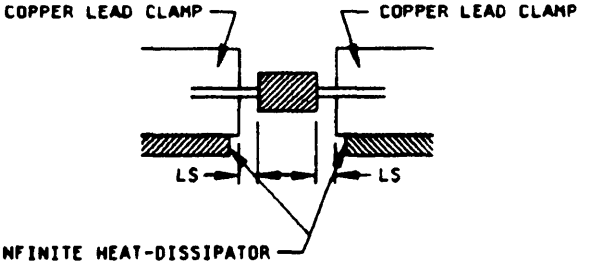
b. t_H . . . . . Thermal equilibrium.

c. I_M . . . . . 1.0 mA to 10 mA.

d. t_{MD} . . . . . 100 µs maximum.

The device shall be allowed to readch equilibrium at current I_H before the measurement shall be made ($t_H \geq 25$ sec).

LS = Lead spacing = 9.53 mm (.375 inch) minimum for leaded devices and LS = 0 minimum for unleaded devices as defined on figure 6 below:



NOTES:
1. Dimensions are in millimeters.
2. Metric equivalents are given for general information only.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.3	$Z_{\theta JX}$		1.5	°C/W
Forward voltage	4011	I_f = 9.0 A(pk); duty cycle ≤ 2% (pulsed); t_p ≤ 8.3 ms	V_{f1}			
1N5550,1N5550US				0.6	1.2	V(pk)
1N5551,1N5551US				0.6	1.2	V(pk)
1N5552,1N5552US				0.6	1.2	V(pk)
1N5553,1N5553US				0.6	1.3	V(pk)
1N5554,1N5554US				0.6	1.3	V(pk)
Forward voltage	4011	I_F = 1.5 A dc	V_{F1}	0.5	1.0	V dc
Reverse current leakage	4016	DC method	I_{R1}			
1N5550,1N5550US		V_R = 200 V dc			1.0	µA dc
1N5551,1N5551US		V_R = 400 V dc			1.0	µA dc
1N5552,1N5552US		V_R = 600 V dc			1.0	µA dc
1N5553,1N5553US		V_R = 800 V dc			1.0	µA dc
1N5554,1N5554US		V_R = 1,000 V dc			1.0	µA dc
Breakdown voltage (diodes)	4021		V_{BR1}			
1N5550,1N5550US		I_R = 50 µA dc		240		V dc
1N5551,1N5551US		I_R = 50 µA dc		460		V dc
1N5552,1N5552US		I_R = 50 µA dc		660		V dc
1N5553,1N5553US		I_R = 50 µA dc		880		V dc
1N5554,1N5554US		I_R = 50 µA dc		1,100		V dc
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5550,1N5550US		V_R = 200 V dc			75	µA dc
1N5551,1N5551US		V_R = 400 V dc			75	µA dc
1N5552,1N5552US		V_R = 600 V dc			75	µA dc
1N5553,1N5553US		V_R = 800 V dc			75	µA dc
1N5554,1N5554US		V_R = 1,000 V dc			75	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Forward voltage	4011	I_f = 9.0 A(pk); duty cycle ≤ 2% (pulsed); t_p ≤ 8.3 ms	V_{f2}			
1N5550,1N5550US					1.2	V(pk)
1N5551,1N5551US					1.2	V(pk)
1N5552,1N5552US					1.2	V(pk)
1N5553,1N5553US					1.3	V(pk)
1N5554,1N5554US					1.3	V(pk)
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	I_f = 9.0 A(pk); duty cycle ≤ 2% (pulsed); t_p ≤ 8.3 ms	V_{f3}		1.5	V(pk)
Forward voltage	4011	I_F = 1.5 A dc	V_F	0.5	1.2	V dc
Breakdown voltage (diodes)	4021		V_{BR2}			
1N5550,1N5550US		I_R = 50 µA dc		200		V dc
1N5551,1N5551US		I_R = 50 µA dc		400		V dc
1N5552,1N5552US		I_R = 50 µA dc		600		V dc
1N5553,1N5553US		I_R = 50 µA dc		800		V dc
1N5554,1N5554US		I_R = 50 µA dc		1,000		V dc
Subgroup 4						
Reverse recovery time	4.31	Condition B-1	t_{rr}		2.0	µs
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1	Reverse current leaking change	4016	DC method	ΔI_{R1} 3/		±100% of initial value or ±250 nA dc whichever is greater.	
2	Forward voltage change	4011	I_F = 1.5 A dc; Pulsed (see 4.5.1)	ΔV_{F1} 3/		±50 mV dc maximum change from previous measured vaule.	

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 2.

b. Subgroup 4, see table II herein, step 2.

c. Subgroup 5, see table II herein, steps 1 and 2.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 1.

b. Subgroup 6, see table II herein, step 1.

3/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Temperature cycling	1051	500 cycles, Condition C	
Electrical measurement 1/		See table I, group A, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state dc blocking life	1038	1,000 hours, condition A $V_R = V_{RWM}$	
Electrical measurement 1/		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3101 or 4081	See 4.5.5; $R_{\theta JL}$ = 22°C/W $R_{\theta JEC}$ = 11°C/W	
Junction temperature rise (see 4.5.4)		See figures 6, 7, and 8; $\Delta T_J \leq 120°C$; L = .375 inch; T_L = 55°C; I_O = 5 A dc 2/	
Subgroup 5			22 devices c = 0
Barometric pressure, reduced (altitude operation)	1001	Pressure (see 1.3); t = 1 min. DC method, $V_R = V_{RWM}$ (see 1.3) I_{R1} = 1.0 µA dc maximum	

1/ $Z_{\theta JX}$ not applicable.
2/ For surface mount device, I_O = 5 A dc and T_{EC} (end cap temperature = 75°C.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Type	V_f at I_f = 9.0 A(pk) 2% duty cycle, 8.3 ms max pulse width		I_{R1}		I_{R2} at T_A = +100°C	
	Min V(pk)	Max V(pk)	µA dc (max) at V_R (V dc)		µA dc (max) at V_R (V dc)	
1N5550,1N5550US	0.6	1.2	1.0	200	75	200
1N5551,1N5551US	0.6	1.2	1.0	400	75	400
1N5552,1N5552US	0.6	1.2	1.0	600	75	600
1N5553,1N5553US	0.6	1.3	1.0	800	75	800
1N5554,1N5554US	0.6	1.3	1.0	1,000	75	1,000

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in this solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used in this specification are defined in MIL-S-19500 and as defined herein.

US . . . . . Unleaded or surface mounted (square encapped diodes).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, 3, 4, and 5.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

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3.3.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins (see MIL-S-19500). Metallurgical bond shall be in accordance with the requirements of category I in MIL-S-19500. US version devices shall be structurally identical to the non-surface mount devices except for lead terminations.

3.4 Marking. Device marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of US version. For US version only, all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

3.4.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. Alternately for surface mount (US) devices, a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used. No color coding will be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table III herein.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500, appendix H.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. JANC die shall be in accordance with MIL-S-19500, appendix H.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	V_{f1} and I_{R1}	Not applicable
11	V_{f1} and I_{R1}; $\Delta V_{f1} \leq \pm 0.1$ V dc $\Delta I_{R1} \leq \pm 250$ nA dc or 100 percent of initial value whichever is greater.	V_{f1} and I_{R1}
12	See 4.3.1.	See 4.3.1.
13 2/	Subgroups 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 250 nA dc, whichever is greater. $\Delta V_{f1} \leq \pm .1$ V dc change from initial value. Scope display evaluation (see 4.5.3).	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 250 nA dc, whichever is greater. $\Delta V_{f1} \leq \pm .1$ V dc change from initial value. Scope display evaluation (see 4.5.3).

1/ Thermal impedance shall be performed any time after screen 3.
2/ Except thermal impedance, if already performed.

4.3.1 Power burn-in test conditions. Power burn-in conditions (all levels) are as follows (see 4.5.1): T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O = 3 A dc.

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4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H.

4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit shall not to exceed the group A, subgroup 2 limit for $Z_{\theta JX}$ in screening (table II of MIL-S-19500).

4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record date $Z_{\theta JX}$). $Z_{\theta JX}$ shall be supplied on one lot (500 pieces minimum and a thermal response curve shall be submitted. Twenty-two of these samples shall be serialized and provided to the qualifying activity for correlation prior to shipment of parts. Measurements conditions shall be in accordance with 4.4.1.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection: $Z_{\theta JX} \leq 1.5°C/W$.

I_M	1 mA to 10 mA.
I_H	5 A minimum.
t_H	10 ms.
t_{MD}	100 µs maximum.
t_{SW}	5 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and table II herein except $Z_{\theta JX}$ shall be performed after intermittent life only.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	4066	I_{FSM} = 100 A(pk), 10 surges of 8.3 ms each at 1-minute intervals, superimposed on I_O = 2 A dc; V_R = rated V_{RWM} (see 1.3); T_A = +55°C. This test shall be performed on each sublot. For mounting conditions, see 4.5.1.1 and 4.5.1.2.
B4	1036	I_O = 3 A dc; f = 50-60 Hz; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. V_R = maximum rated V_{RWM} (see 1.3 and 4.5.1); $t_{on} = t_{off}$ = 3 minutes minimum for 2000 cycles.
B5	1027	T_A = +150°C minimum, I_O = 3 A dc minimum. Adjust T_A or I_O as required by the corresponding chosen value of T_A or I_O to achieve a lot T_J = +275°C. Delta limits: $\Delta I_R \leq$ 100 percent or 50 nA, whichever is greater.
B6	4081 or 3101	T_L = +25°C to +35°C; $R_{\theta JL}$ = 22°C/W maximum.L = .375 inch; $R_{\theta J}$ endcap = 11°C/W maximum (US version).
B7		Peak reverse power: See figure 9 and 4.5.6; PRM = 1000 watts (minimum). This test shall be performed on each sublot. LTPD = 10.

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4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B2	4066	I_{FSM} = 100 A(pk), 10 surges of 8.3 ms each at 1-minute intervals, superimposed on I_O = 2 A dc; V_R = rated V_{RWM} (see 1.3); T_A = +55°C. This test shall be performed on each sublot. For mounting conditions, see 4.5.1.1 and 4.5.1.2.
B3	1027	I_O = 3 A dc; f = 50-60 Hz; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. V_R = rated V_{RWM} (see 1.3 and 4.5.1). For mounting conditions see 4.5.1.1 and 4.5.1.2.
B7		Peak reverse power: See figures 9 and 4.5.6; PRM = 1000 watts (minimum). This test shall be performed on each sublot. LTPD = 10.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable inspections of table I, group A, subgroup 2.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	1056	Test Condition B
C2	2036	Test Condition A Tension: Weight = 5 pounds; t = 15 seconds Fatigue: Test condition E; weight = 2 pounds NOTE: Terminal strength not applicable for US version devices.
C2	1021	Omit initial conditioning.
C5		Not applicable.
C6	1026	I_O = 3 A dc; f = 50-60 Hz; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. V_R = rated V_{RWM} (see 1.3 and 4.5.1); for mounting conditions, see 4.5.1.1 and 4.5.1.2.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Steady-state operation life. A half-sine wave of the specified peak voltage shall be impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

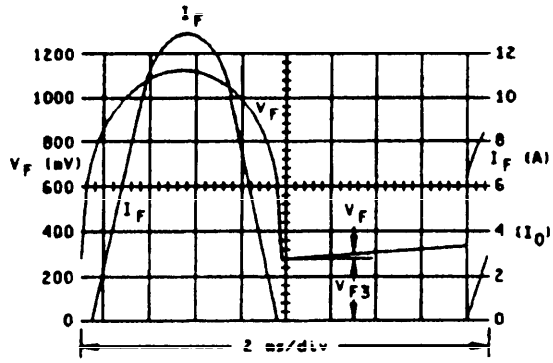
4.5.1.1 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_O is increased so that the junction temperature of each diode is maintained at +145°C minimum.

4.5.1.2 Alternate mounting conditions (for -US devices). At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that one of the following conditions be met:

a. T_{EC} = +75°C to +125°C, V_{RWM} = rated, f = 50-60 Hz, I_O = 3 amps.

b. Temporary attachment of leads or equivalent (thermal properties not to exceed the leaded part) f = 50-60 Hz. I_O = 3 amps, T_J = +145°C minimum, V_R = rated V_{RWM}.

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NOTE: Blocking diode shall have a forward current rating ≥ 6 A dc.

FIGURE 6. Junction temperature rise test circuit.

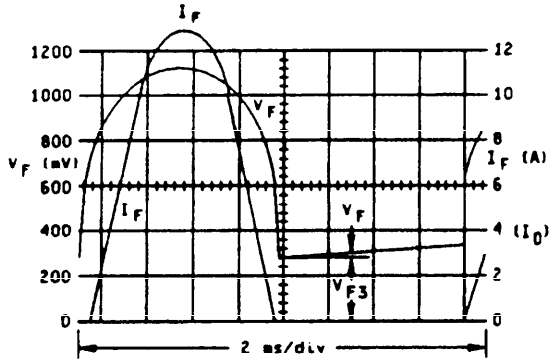


FIGURE 7. Junction temperature test oxcillogram (typical).

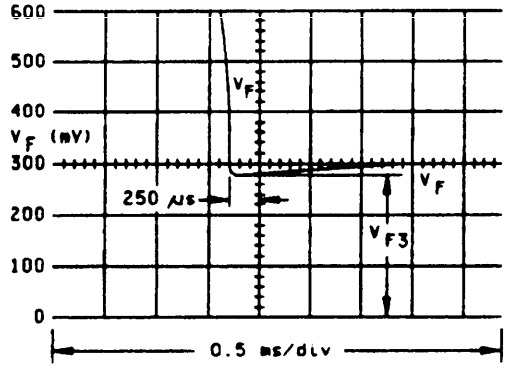
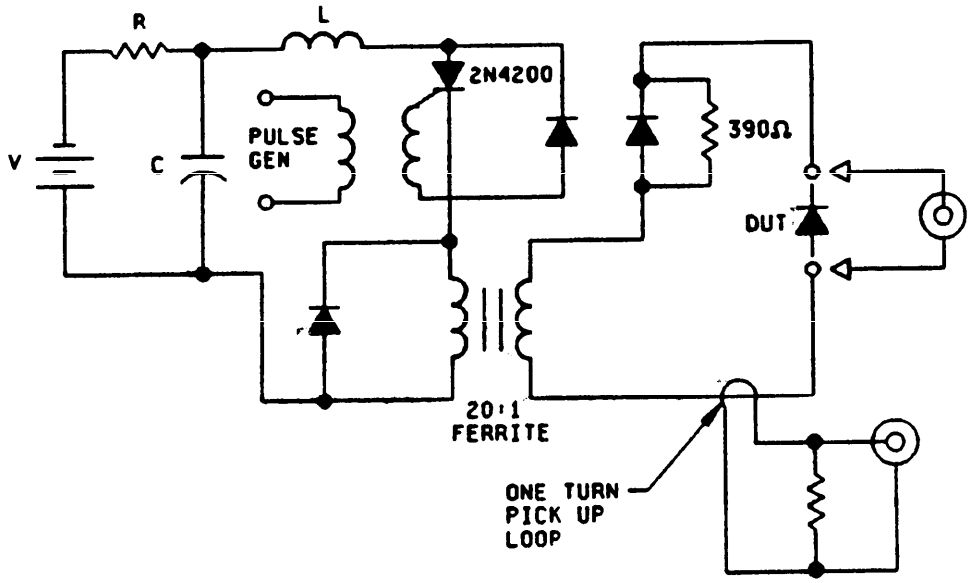
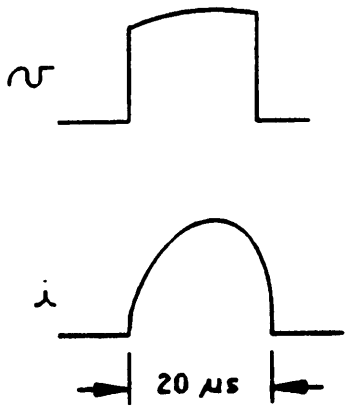


FIGURE 8. Expanded oscillogram of V_F.

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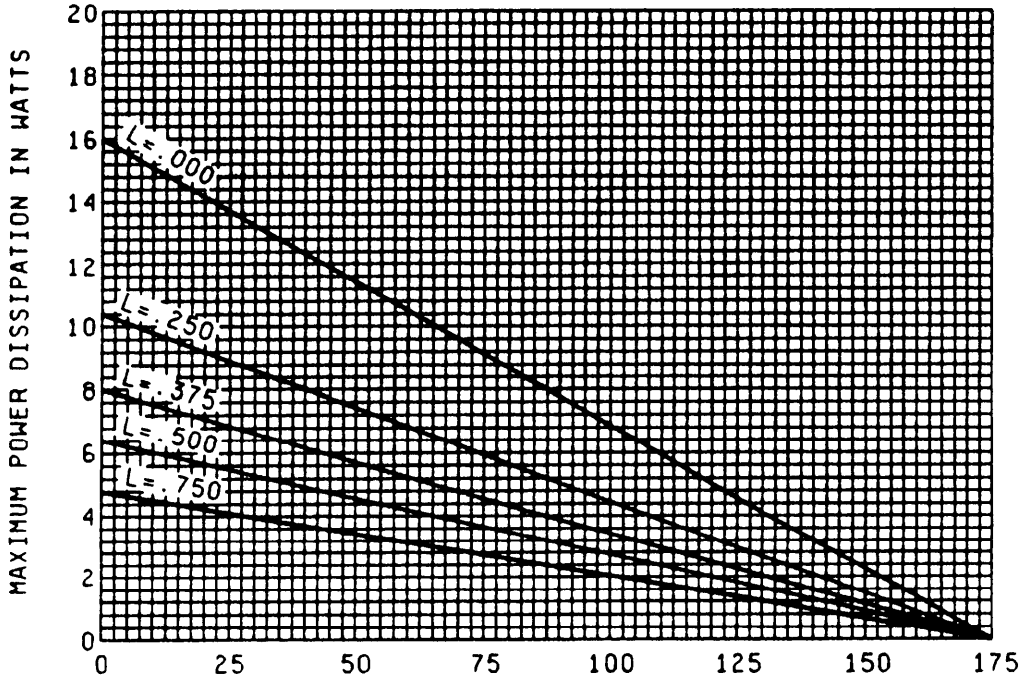
NOTES:
1. L = 13T #22 on 1 inch diameter form (air core).
2. C = 1 to 10 μF to give 20 μs pulse width.
3. V = Adjustable to 200 volts for power desired in device under test.



TYPICAL WAVEFORMS

FIGURE 9. Typical peak reverse power measurement circuit and waveforms.

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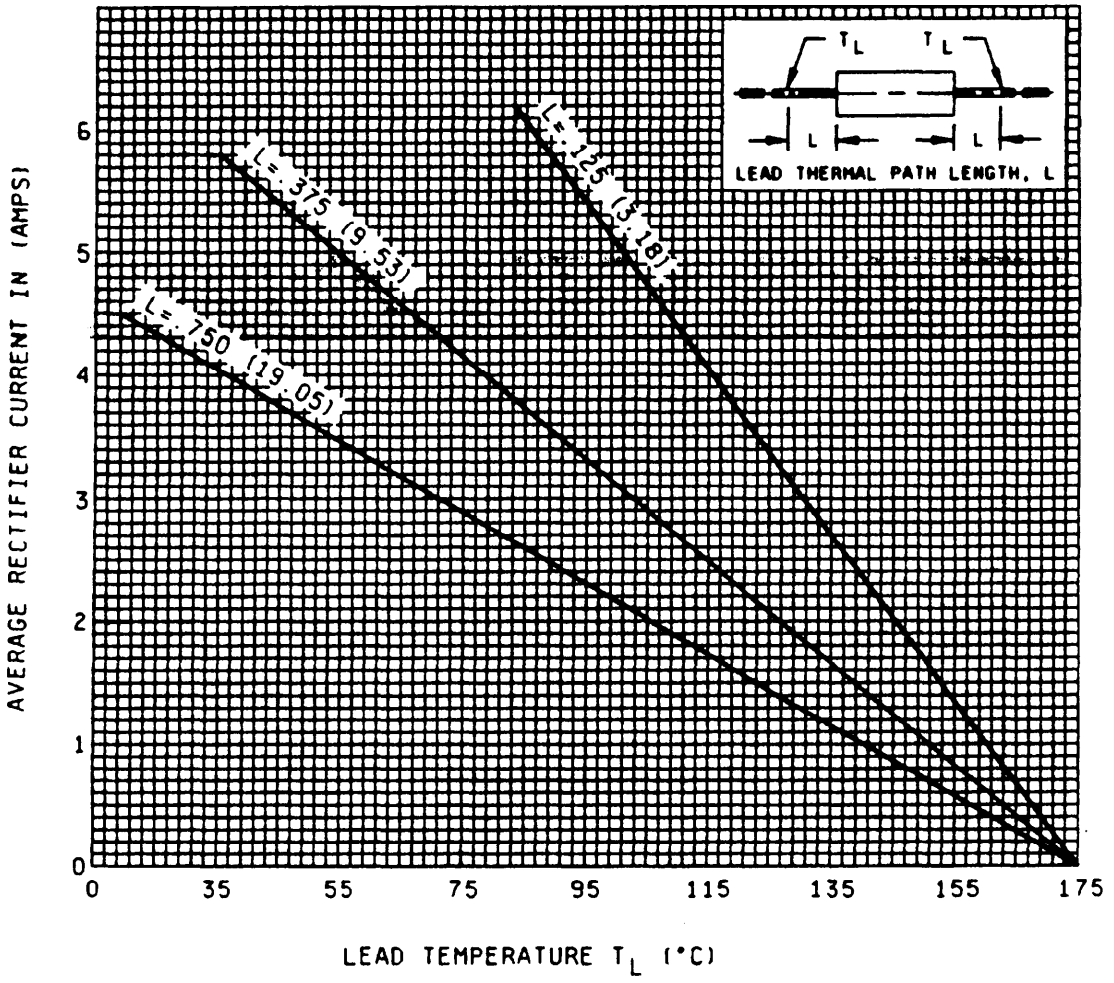
Maximum lead temperature in °C (T_L) at point "L" from body (for maximum operating junction temperature of +175°C with equal two-lead conditions).

L		$R_{\theta JL}$
Inches	mm	°C/W
.000		11
.250	6.35	16.5
.375	9.53	22
.500	12.70	26
.750	19.05	35.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 10. Maximum power in watts vs lead temperature.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 11. Maximum current vs lead temperature.

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4.5.6 Peak reverse power test. This test shall be measured in the circuit on figure 7 or equivalent. A 20 microsecond half-sine wave of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Product assurance level, type designator, and for die acquisition, the JANHC and JANKC identification (see figures 3, 4, and 5 and see 6.5).

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that the manufacturers' PIN's are suitable as a substitute for the military PIN.

6.4 Applications data. Device current capability with lead-dissipators or body forced-air- cooling, may be determined from figure 11, which shows maximum average rectified current versus lead temperature as a function of the distance L from the diode body at which lead temperature is measured. See figure 10 for maximum power in watts as a function of lead temperature at a distance "L" from the diode body.

6.5 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N5550) will be identified on the QPL.

JANC ordering information

PIN	Manufacturer		
	14552	60211	13409
1N5550	JANHCA1N5550 JANKCA1N5550	JANHCB1N5550	JANHCC1N5550
1N5551	JANHCA1N5551 JANKCA1N5551	JANHCB1N5551	JANHCC1N5551
1N5552	JANHCA1N5552 JANKCA1N5552	JANHCB1N5552	JANHCC1N5552
1N5553	JANHCA1N5553 JANKCA1N5553	JANHCB1N5553	JANHCC1N5553
1N5554	JANHCA1N5554 JANKCA1N5554	JANHCB1N5554	JANHCC1N5554

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – ER
Navy – EC
Air Force – 17
NASA – NA

Review Activities:
Army – AR, MI, SM
Navy – AS, CG, MC
Air Force – 19, 80, 85, 99
DLA– ES

Preparing Activity:
NASA-Na

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-S-19500/420D	**2. DOCUMENT DATE (YYMMDD)** 94-07-22

DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER, RECTIFIER, GENERAL PURPOSE, TYPES 1N5550 THROUGH 1N5554, 1N555OUS THROUGH 1N5554US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ALAN BARONE	b. TELEPHONE *Include Area Code)* (1) Commercial (513-296-6048) (2) AUTOVON 986-6048
c. ADDRESS *(Include Zip Code)Commander* DEFENSE ELECTRONICS SUPPLY CENTER ATTN: DESC-ELDT DAYTON, OH 45444-5765	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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